

CARRIERS, DEVICES AND FRESH ROOTS: THE ARROW METHOD FROM MINIMUM MODULUS 42 TO 44

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ABSTRACT. Nielsen and Owens constructed covering systems with distinct moduli whose smallest moduli are 40 and 42. They filled “holes” prime by prime with arrows. We describe the ideas that carry this method to 44. A capacity bound decides which arrows can fit on a given piece of a hole. Carriers, congruences that cover whole slots of arrows in other sections, can be routed to where they are needed by relabelling them modulo 5. The unique family of smallest modulus can be cut back to its deeper levels. The part of the integers that its first level alone covered is then small, and an exact computation lists where it lies. We cover it again with roots on fresh primes, helped by a small device on the prime 7 whose digits serve as exact carries. For the second step we nest a smaller arrow into the slot of the offending input. The thin remainder is closed by further roots and by local repairs found from exact counterexamples.

1. INTRODUCTION

A finite set of congruence classes $a_i \bmod m_i$ with distinct moduli $m_i > 1$ whose union is $\mathbb{Z}$ is a *covering system*. In 1950 Erdős [5] asked whether the smallest modulus of a covering system can be arbitrarily large. Filaseta, Ford, Konyagin, Pomerance and Yu [6] restricted such systems severely, and Hough [8] proved that the answer is no: the smallest modulus is at most 10^{16} . Balister, Bollobás, Morris, Sahasrabudhe and Tiba [1] lowered the bound to 616,000.

Constructions have come slowly. Churchhouse [4] found a covering system with smallest modulus 9 in 1968. In 1971 Krukenberg [9] reached 18 and Choi [3] 20. About ten years later Morikawa [11, 12] reached 24, and Gibson [7] then reached 25. Nielsen [13] introduced a compact notation for such constructions, built around *arrows*, and reached 40. Owens [14] extended the method to 42.

Theorem 1.1. *There is a covering system with distinct moduli whose smallest modulus is 44.*

The system, 618,413 families of congruences on the 47 primes up to 211, is given with its verification in the companion paper [10]. Here we explain how it was found, in the manner of [13] and [14]: which hole we are filling, which sets are available, and how many sets we obtain. We start from a system with smallest modulus 42, a variant of Owens’s (§4). We reach 43 in §§5–7 and 44 in §8. As far as we know, the following ideas are new.

- *A capacity bound* (§3.1). It bounds how many pairwise separated covers a piece of the integers can hold, and so which primes can root an arrow there.
- *Exact censuses* (§3.2). Whether a region is covered is a finite question. A counterexample loop lists exactly where a hole lies and certifies that the list is complete.
- *Carrier routing* (§4). A few congruences cover whole slots of arrows in other sections. Relabelling such a carrier modulo 5 decides which sections receive its carry.
- *Cutting back the smallest family* (§5). The unique family of smallest modulus keeps its deeper levels and gives up only its first.
- *Devices* (§6). A few families on deep rays cover fixed digits exactly, and so serve as exact carries.

- *Fresh roots* (§7). An arrow on a prime used nowhere else covers one piece of the hole. Its inputs are separated covers of that piece.
- *Nesting in the offending slot* (§8). The input responsible for the smallest modulus is replaced by an arrow on a smaller prime. Every other input stays in its slot.
- *Local repairs* (§8). Each uncovered point that the exact computation returns is traced to the input that misses it, and that input is patched in place.

2. NOTATION

We follow Nielsen [13, §2]. For a prime q , the expression $q(A_1, \dots, A_q)$ denotes the congruences obtained by placing the set A_j on the branch $j \bmod q$. Each set is scaled by q : an entry m on branch j stands for a congruence class of modulus qm inside $j \bmod q$. The *arrow* is the infinite recursion

$$q\uparrow(A_1, \dots, A_{q-1}) = q(A_1, \dots, A_{q-1}, q\uparrow(A_1, \dots, A_{q-1})).$$

The same $q - 1$ *inputs* are reused on every level, deeper and deeper inside $0 \bmod q$. The input A_j thus covers, inside the region, the integers $x \neq 0$ whose first nonzero digit in base q is j . We call this set the *slot* j of the arrow, and the digit its *q -lead*. For example, in $3\uparrow(1, 2\uparrow)$ the first input contributes the class $3^{k-1} \bmod 3^k$ on every level k . The second contributes $2 \cdot 3^{k-1} \bmod 3^k$, intersected with the congruences of $2\uparrow$ [13, Example 1]. Nielsen stops the recursion at a deep level and covers what is left with the help of a large auxiliary prime. At a deep enough level every modulus this introduces is large, and distinct powers keep the moduli distinct.

For bookkeeping we write the congruences produced by one path through the arrows as a *family*: a product over primes of *coordinates*. A coordinate is either *fixed*, one class $r \bmod p^e$, or a *ray* $\text{ray}(p, [\ell, h], c, d)$: the classes $c + dp^{e-1} \bmod p^e$ for $\ell \leq e \leq h$ (we allow $h = \infty$). The number c is the ray's *centre*. A family's moduli are the products of one prime power per coordinate, and its *box* is the product of its exponent ranges. Two families can repeat a modulus only if they involve the same primes and their boxes overlap. When no two families of a collection do this, we call the collection *separated*.

Two conventions of Owens [14] matter below. First, an input of multiplier μ in a q -arrow gives a first-level modulus $q\mu$. It is therefore allowed only if $q\mu \geq N$, where N is the target smallest modulus. For $N = 42$, for instance, the inputs 1 and 2 must be dropped from a 19-arrow. Second, a slot is *carried* where earlier material already covers it, and a carried slot needs no input there.

We refer to the sections of the construction by their numbers in [14]. For instance, §3.4 is the section of the prime 7, §3.5 that of 11 and §3.14 that of 43. A *cell* is a set of the form $\{x : v_2(x) = k, x \equiv r \bmod 5\}$ inside a fixed residue class, written (k, r) . A *piece* is a cell further restricted by one or more leading digits.

3. TWO TOOLS

3.1. A capacity bound. Much of what follows is a question of whether a piece of the integers has room for $q - 1$ separated covers. The following bound answers it in the negative surprisingly often.

Lemma 3.1. *Let Q be a finite set of primes. Let R be a residue class modulo $\prod_{p \in Q} p^{a_p}$, where $a_p \geq 0$. Let $T_1, \dots, T_n$ be sets of congruence classes with Q -smooth moduli, each covering R , and suppose that no modulus occurs twice among all of them. Then*

$$n \leq \nu(R) := \prod_{p \in Q} \left(a_p + 1 + \frac{1}{p-1} \right).$$

If every modulus is at least t , then more precisely $n \leq \sum_{m \geq t} \prod_{p \in Q} \min(1, p^{a_p - v_p(m)})$, the sum running over the Q -smooth $m \geq t$.

Proof. A class of modulus m meets R in a class modulo $\text{lcm}(m, \prod p^{a_p})$, or not at all. Its share of R is therefore at most $w(m) = \prod_p \min(1, p^{a_p - v_p(m)})$. Since T_i covers R , the shares of its classes add up to at least 1. Summing over i and using that the moduli are distinct gives $n \leq \sum_m w(m)$. Without the restriction $m \geq t$ the sum factors as $\prod_p \sum_{e \geq 0} \min(1, p^{a_p - e}) = \prod_p (a_p + 1 + \frac{1}{p-1})$. $\square$

An unpinned prime contributes the factor $p/(p-1)$. A computation of the same kind shows that a piece pinned by a single leading digit at p , on every level, gets the factor 2. A root on q over R with c carried slots needs $q-1-c$ inputs that cover R and do not repeat each other's moduli once the factor q is removed. So it fits only if $q-1-c \leq \nu(R)$, computed without the prime q . The bound is also nearly attained in practice.

Remark 3.2. Measured against this bound, the construction of Owens is nearly tight. Each of its sections needs between 93% and 100% of ν for its root (median about 96%). Our starting system spends 94.7% of the sum 3.830 of $1/m$ over the 83-smooth $m \geq 42$. So every gain has to come from new room: fresh primes, or lattice points that no family uses.

3.2. Exact censuses. Everything below depends on knowing exactly where a hole is. This is possible because coverage is a finite question.

Lemma 3.3. *Let F be a finite set of families. For each prime p involved there is a partition of $\mathbb{Z}$ into finitely many atoms such that every coordinate of F on p is a union of atoms, and every atom is either a single ray centre or contains a full residue class modulo a power of p . Hence whether an integer is covered by F depends only on the atoms that contain it. Every choice of one atom per prime, none of them a centre, is realised by infinitely many integers.*

Proof. Take E larger than every finite exponent in F and than $v_p(c-c')$ for any two distinct centres c, c' on p . The atoms are the classes modulo p^E that contain no centre. The class of each centre c is split into the point c and, for $j = 1, \dots, p-1$, the integers $x \neq c$ in it whose leading digit of $x-c$ is j . A fixed coordinate of exponent less than E is a union of classes modulo p^E . A ray $\text{ray}(p, [\ell, h], c, d)$ consists of the x with $\ell-1 \leq v_p(x-c) \leq h-1$ and leading digit d , so it is a union of atoms as well. The atom with leading digit j contains the class $c + jp^E \bmod p^{E+1}$. The last claim follows from the Chinese remainder theorem. $\square$

So whether a region X is covered is the satisfiability question for a propositional formula. It has one variable per atom and one clause per family, which says that the point is not in that family. It also has clauses saying that the point takes exactly one atom per prime and lies in X . The formula is unsatisfiable if and only if X is covered, and a solver such as CaDiCaL [2] gives a proof of unsatisfiability that can be checked independently [15]. A satisfying assignment that avoids the centres decodes into an explicit uncovered integer.

We use this in two loops. In a *census*, each counterexample is widened to the piece that contains it: its atoms at a few small primes, and its leading digit at the root prime of the section that owns it. That piece is blocked, and we ask again. When the formula becomes unsatisfiable, the blocked pieces contain the whole hole, and the final proof certifies that the list is complete. In a *repair*, each counterexample is instead traced to the input that should have covered it, that input is patched, and we ask again (§8).

4. CARRIER ROUTING AND THE STARTING SYSTEM

In the sections for 11, 13 and 17 (§§3.5–3.7) a few bare congruences do more than their own job. They cover a whole slot of an arrow used much later, on an even branch. We call these

congruences *carriers*. Each one lies in a single class modulo 5, and that class is a digit of its own 5-arrow. The later sections live in different classes modulo 5, so the label of a carrier decides which of them receive its carry. The three that carry a whole slot are listed in Table 1, where $q:j$ is the slot of digit j in the q -arrows of the receiving sections.

TABLE 1. The whole-slot carriers of §§3.5–3.7 and the later sections that receive their carry, by label.

carrier	modulus	slot	label	sections receiving the carry
C_{55}	55	11:6	1	§§3.8, 3.12, 3.17
			4	§§3.14, 3.15, 3.16, 3.18, 3.20
C_{65}	65	13:6	2	§§3.10, 3.13, 3.19
			4	§§3.14, 3.15, 3.16, 3.18, 3.20
C_{85}	85	17:9	1	§§3.8, 3.12, 3.17
			4	§§3.14, 3.15, 3.16, 3.18, 3.20

A carried slot is worth one whole input to the arrow that receives it, so moving a carrier moves one input from one section to another. Here is how we used this. Give C_{55} , C_{65} and C_{85} the labels 1, 2 and 1. The 67-arrow of §3.18 then finds only 63 of the 64 sets of its own that it needs. The other two of its sixty-six inputs come, as in [14], from doubling. There are two ways to give that arrow a whole slot.

- *Move C_{65} to label 4.* The 67-arrow then reaches 64. But §3.19 uses 13-arrows with only eleven inputs, because their slot 6 is carried there. Without the carry, both halves of §3.19 fall eight inputs short.
- *Move C_{85} to label 4.* The 67-arrow again reaches 64. The sections that lose the carry do not need it: §3.8 still builds its twenty sets, and §§3.12 and 3.17 still close.

So we move C_{85} . Concretely, its 5-arrow in §3.7 is written $5\uparrow(8\uparrow, 2, 4, 1)$ instead of $5\uparrow(1, 2, 4, 8\uparrow)$. The remaining arrangement is done by an exact planner. For each section, a dynamic programme over subsets of the available base sets chooses the order in which the arrows are built. It re-plans §§3.12, 3.14 and 3.15 and both arrows of §3.18, and §3.18 needs no 89-arrow. We call the result UC85. It has 356,364 families on the 23 primes up to 83, and its smallest modulus 42 is attained exactly once.

5. CUTTING BACK THE SMALLEST FAMILY

The modulus $42 = 2 \cdot 3 \cdot 7$ comes from one family of §3.4,

$$F_{42} = \{0 \bmod 2\} \cdot \{1 \bmod 3\} \cdot \text{ray}(7, [1, \infty), 0, 3),$$

whose first level is the class $10 \bmod 42$. To reach 43 we start its ray one level lower, at $\text{ray}(7, [2, \infty), 0, 3)$. The moduli $2 \cdot 3 \cdot 7^e$ with $e \geq 2$ remain, and only the class $10 \bmod 42$ is given up (Figure 1).

Most of $10 \bmod 42$ is still covered by other families. What is left, the hole U , is about 2.1% of the class, or $4.95 \cdot 10^{-4}$ of the integers. Every integer in $10 \bmod 42$ is even and $\equiv 3 \bmod 7$. Sorting by $v_2(x)$ and $x \bmod 5$, an exact census (§3.2) shows that U meets exactly fourteen cells. Table 2 lists them with their sizes, which were estimated by sampling.

The hole has two sources. Where $v_2(x) \geq 2$ it is mostly the part of $10 \bmod 42$ that F_{42} alone covered. The row $v_2 = 1$ and the cells $(2, 1)$ and $(4, 4)$ come from carries that F_{42} had supplied to arrows of later sections: §§3.16, 3.8, 3.10, 3.10 and 3.15 for $(1, 0), \dots, (1, 4)$, §3.12 for $(2, 1)$ and the 79-arrow of §3.20 for $(4, 4)$. For instance, on the cell $(1, 1)$ the hole lies only in the

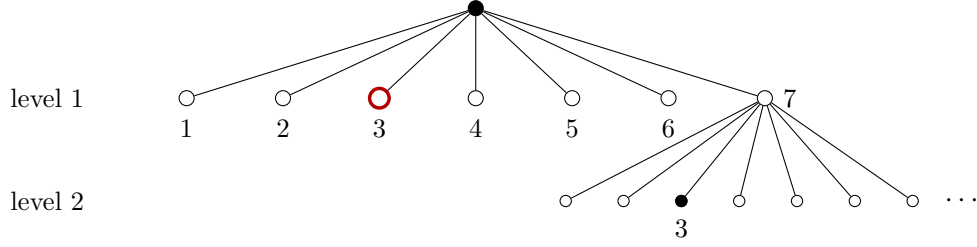


FIGURE 1. The 7-tree of F_{42} on the branch $\{0 \bmod 2\} \cap \{1 \bmod 3\}$. Branch 7 is $0 \bmod 7$, where the tree recurs. The ray covers the digit 3 on every level. Cutting it back gives up only the outlined first-level class, $10 \bmod 42$; the filled class and its copies further down are kept.

TABLE 2. The hole U after cutting back F_{42} : the uncovered fraction of each cell $(v_2(x), x \bmod 5)$ of $10 \bmod 42$ (sampled). A dash marks a cell that U does not meet; U also contains no multiple of 32.

$v_2 \setminus x \bmod 5$	0	1	2	3	4
1	0.14%	0.65%	0.80%	1.58%	0.62%
2	0.41%	2.21%	6.65%	8.66%	5.36%
3	—	—	7.85%	10.2%	5.40%
4	—	—	—	—	2.49%

19-slots 16, 17 and 18. These are the three copies of $7 \uparrow$ in §3.8, whose slot 3 was carried by F_{42} . The holes in $(1, 0)$ and $(2, 0)$ lie in $20 \bmod 25$.

One might hope to re-plan the sections that lose a carry. This fails. With F_{42} cut back, §3.8 reaches 19 of its 20 sets, §3.10 27 of 28, §3.12 34 of 36, §3.15 44 of 46 and §3.16 48 of 52. The 67-arrow of §3.18 falls to 65 of 66. A section that is short leaves a whole root slot open on its whole region, which is far worse than U . So we keep the sections as they are and cover U from outside.

6. A DEVICE ON THE PRIME 7

The whole hole lies in $3 \bmod 7$, where 7 divides $x + 4$. Write $R_j = \text{ray}(7, [2, \infty), -4, j)$. Inside $3 \bmod 7$, R_j covers exactly the integers $x \neq -4$ for which the first nonzero digit of $x + 4$ in base 7 is j . We use the seven families

$$\begin{aligned} &\{0 \bmod 2\}R_1, \quad \{1 \bmod 3\}R_2, \quad \{0 \bmod 4\}R_3, \quad \{0 \bmod 4\}\{13 \bmod 27\}R_4, \\ &R_5, \quad \{3 \bmod 5\}R_6, \quad \{24 \bmod 32\}\{1 \bmod 3\}\{2 \bmod 5\}R_6. \end{aligned}$$

Their smallest moduli are 98, 147, 196, 5,292, 49, 245 and 23,520. No other family of the system has the same primes and an overlapping box, so they cost nothing. The ray R_5 , for instance, uses the pure powers 7^e , $e \geq 2$, which no other family uses. Inside $10 \bmod 42$, which is even and $\equiv 1 \bmod 3$, the device covers the leading digits 1, 2 and 5 of $x + 4$ everywhere. It also covers the digit 3 on $0 \bmod 4$, the digit 6 on $3 \bmod 5$, and a little more.

We call such a small set of families a *device*. By itself it covers about 70% of U but empties no cell, since every cell contains every digit. Its value is that it is exact and uniform. Any cover of a piece that works deep in the 7-adic direction, with centre -4 , may treat the device's digits as carried slots. On a cell with $v_2 = 1$ these are the digits 1, 2, 5. With $v_2 \geq 2$ the digit 3 is added, and on $x \equiv 3 \bmod 5$ the digit 6.

7. FRESH ROOTS

Suppose a piece P of the hole is to be covered by a q -arrow on a prime q used nowhere else in the system. We call this a *fresh root*. Every congruence of the root carries a power of q , so the root cannot repeat a modulus of any family outside it. It needs $q - 1$ inputs, each a set of congruences covering P , and the inputs must be separated from one another. By Lemma 3.1 this is possible only if $q - 1 \leq \nu(P)$, less the carried slots. We call the largest number of separated covers of P that we can build its *pool*. The building blocks are fixed coordinates, deep rays on the pinned primes, and arrows on the primes from 11 upwards, other than q .

Carried slots raise the pool, but only if they are carried exactly. We use two sources of exact carries: the digits of the device, and the slots of a section's root that an exact computation proves covered on the piece. Carries observed only on samples are not safe. On one cell with $v_2 = 1$ they suggested a pool of 156, and the exact computation showed that they did not hold. Table 3 shows the pools we reached.

TABLE 3. Pools of separated covers. *Base*: covers made of fixed coordinates and rays only. *Pool*: with arrows. ν : the bound of Lemma 3.1 with the primes up to 200. The last column uses exact carries.

piece	base	pool	ν	with exact carries
cell, $v_2 = 1$	46	74	≈ 105	88–96 (root digits, device); 122 split mod 8
cell, $v_2 = 2$	58	104	≈ 133	131 (device digits 1, 2, 3, 5); 140 on (2, 1)
cell, $v_2 = 3$	70	121	≈ 160	at least 148 (device)
cell, $v_2 = 4$	83	160	≈ 186	–
(1, 1), a 19-digit pinned	95	177	≈ 202	–
(1, 0), (2, 0) in 20 mod 25	68	$\geq 126, \geq 150$	–	–

The primes 11 to 83, which the construction already uses, cannot serve as roots. The sections that use them occupy the low-lying lattice points, and a root on one of them would have room for at most seven covers. Two principles then decide the assignment. First, the weakest cells, those with $v_2 = 1$, go on the smallest primes, since a root on q needs only $q - 1$ inputs. Second, a cell whose pool falls just short is split, by $x \bmod 8$ or by a leading digit, and each part gets a root of its own. Our first assignment, with twelve roots, the device and the fills described below, left 48 pieces uncovered. All of them lay in the cells (1, 2), (1, 3), (1, 4) and (2, 1), at a few leading digits of the roots of §§3.10, 3.15 and 3.12. The same computation proves the other digits of those roots covered on these cells. Used as exact carries, they are what made the final assignment of seventeen roots work, and the final system is checked again as a whole.

The Prime 89. We begin with the cell (1, 2): the part of U with $x \equiv 2 \bmod 4$ and $x \equiv 2 \bmod 5$. It lies in the region of §3.10, whose 29-root covers nineteen of its twenty-eight slots on this cell (an exact computation shows this). The device gives the 7-digits 1, 2 and 5. With these carries we obtain eighty-eight sets, which is exactly what $89\uparrow$ needs (5,818 families).

The Primes 97, 101 and 103. The cell (1, 3) is handled in the same way, now also with the device digit 6, since $x \equiv 3 \bmod 5$ there: ninety-six sets for $97\uparrow$ (16,972 families). The cell (1, 4) lies in the region of §3.15. Its 47-root covers twenty-eight slots there, but even with that carry the cell is too large for one root. We split it by $x \bmod 8$, covering $2 \bmod 8$ with $101\uparrow$ and $6 \bmod 8$ with $103\uparrow$.

The Primes 107 through 173. The remaining cells are filled one root at a time, as listed in Table 4. The cells with $v_2 \geq 2$ use the device digit 3 as well. The cell $(1, 1)$, whose hole lies in three 19-slots, is pinned further by its 19-digit, and each digit gets a root of its own.

TABLE 4. The seventeen fresh roots of the step to 43. A cell $(v_2(x), x \bmod 5)$ is taken inside $10 \bmod 42$; “device” lists the 7-digits used as carries.

q	piece	exact carries used	families
89	(1, 2)	29-root, 19 slots; device 1, 2, 5	5,818
97	(1, 3)	29-root, 19 slots; device 1, 2, 5, 6	16,972
101, 103	(1, 4) split by $x \bmod 8$	47-root, 28 slots; device 1, 2, 5	3,280, 4,728
107	(1, 0) in $20 \bmod 25$	device 1, 2, 5	2,052
109, 113	(2, 2), (2, 4)	device 1, 2, 3, 5	4,508, 7,898
127, 149	(2, 3), (3, 3)	device 1, 2, 3, 5, 6	6,181, 3,006
131	(2, 1)	37-root, 24 slots; device 1, 2, 3, 5	33,806
137, 139	(3, 2), (3, 4)	device 1, 2, 3, 5	13,940, 22,278
151, 157	(2, 0) in $20 \bmod 25$, (4, 4)	device 1, 2, 3, 5	4,456, 8,496
163, 167, 173	(1, 1), 19-digits 16, 17, 18	device 1, 2, 5	7,598, 11,778, 27,056

Filling slots in place. Cutting back F_{42} also leaves some copies of arrows in §§3.10, 3.12 and 3.15 with an empty slot. Where the lattice had room, we filled such a slot directly. We used the slot’s context times the class $1 \bmod 3$ (and $3 \bmod 7$ when needed), with a coordinate on 2 or 5 added when that made the modulus new. This adds 3, 75 and 526 families to the three sections. It helps, but most contexts are already saturated by the sections’ own base rays ($8\uparrow$, $9\uparrow$ and $25\uparrow$), so fills alone cannot replace the roots.

With the device, the fills and the seventeen roots, the class $10 \bmod 42$ is covered again. The system now has 540,826 families on the 40 primes up to 173. Its smallest modulus 43 is attained once, by the input 1 of the 43-arrow of §3.14, and each of the moduli 43 to 48 occurs once.

8. FROM 43 TO 44

The modulus 43 comes from the input of multiplier one of the root $43\uparrow$ of §3.14. This is the bare family $\text{ray}(43, [1, \infty), 0, 1)$, which covers the 43-lead 1 on all of $\mathbb{Z}$. We cut it back to $\text{ray}(43, [2, \infty), 0, 1)$, whose smallest modulus is $43^2 = 1849$. This gives up exactly the class $1 \bmod 43$.

A smaller arrow in the slot. On the region of §3.14 the slot 1 must be covered again, now with moduli at least 44. We re-plan §3.14 with 41 allowed as an inner prime. The planner then builds, among its sets, a 41-arrow covering the region. Its 177 families all involve both 41 and 43, and the smallest modulus among them is $2 \cdot 41 \cdot 43 = 3526$. We put this arrow into slot 1 and keep every other input of $43\uparrow$ where it was:

$$43\uparrow(41\uparrow(\dots), A_2, \dots, A_{42}).$$

Keeping the slots matters. Re-planning §3.14 from scratch also closes §3.14, but it moves inputs to other slots, and later sections that relied on the carries of particular 43-slots break. Figure 2 shows the result.

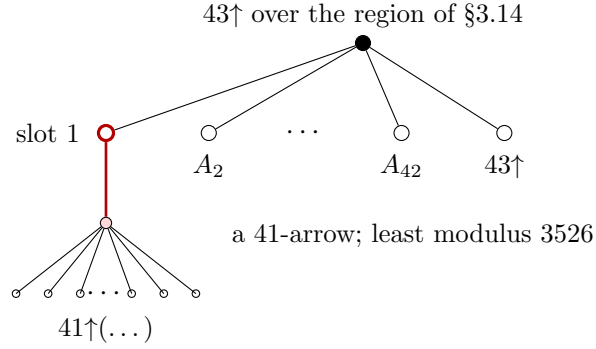


FIGURE 2. The root of §3.14 after the change. Only the input in slot 1 is replaced. Its first level now holds a 41-arrow over the region, and the family of modulus 43 keeps only its deeper levels.

Refilling slot 1 elsewhere. The family we cut back had also carried slot 1 of every copy of $43\uparrow$ nested inside later sections. There are 264 such copies: 2 in §3.15, 1 in §3.16, 3 in §3.17, 76 and 152 in the 61- and 67-arrows of §3.18, and 15 in each of the 71- and 73-arrows of §3.19. We fill each one with the first set that is an input of some arrow of the same section, is not already an input of that copy, and repeats no modulus of the system. This adds 1,183 families. One more copy, at input 80 of the 83-arrow of §3.20, takes a set of two families.

These fills cover what the removed family carried explicitly, but not everything it carried implicitly. Several inputs of the 67-arrow of §3.18 and the 73-arrow of §3.19 are sets that are complete only because some of their own slots are carried, and inside $1 \bmod 43$ those slots had been carried by the removed family. An exact census lists what remains: 1,282 pieces, about $9.5 \cdot 10^{-9}$ of the integers. They lie in sixteen 67-slots (34, 35, 36, 39, 50, 51, 53, 56–60, 63–66) and six 73-slots (57, 62, 63, 66, 69, 70), all in $6 \bmod 8$ and $0 \bmod 3$. None of them lies in the fresh roots of §7.

The Primes 179 through 211. We first ran the local repairs described below. In seven of the 67-slots they ran out of room, and each of these slots gets a fresh root over the piece $54 \bmod 120 \cap 1 \bmod 43 \cap \{67\text{-lead } j\}$. Such a piece holds 254 separated covers (measured at 179), enough for any root up to 211. We use the seven primes from 179 to 211 (Table 5).

TABLE 5. The seven fresh roots of the step to 44.

root prime q	179	181	191	193	197	199	211
67-slot j	34	50	60	39	36	66	63
families	3,312	3,996	6,484	9,772	13,732	13,812	23,316

Local repairs. Everything else is closed by the repair loop of §3.2. The target is $1 \bmod 43$, and the cover is the whole system with the repairs made so far. Each counterexample is traced through the arrows to the input position that should have covered it. That position then receives the first of two kinds of patch that contains the counterexample and repeats no modulus.

- (1) A *thin family*: the position's context times the class $1 \bmod 43$ and the pins of the region.
- (2) A *local arrow* $r\uparrow$ on a prime r that the system already uses, placed inside the context. Every one of its families carries the whole context, including the 43-lead and the 67- or 73-lead of the position. So only families involving r and all of the context's primes can collide with it, and deep inside a context there is room even for $r = 11$.

The loop ended after 110 patches, when the formula became unsatisfiable. They were 15 thin families and 95 local arrows: 89 on the prime 11, two each on 23 and 41, and one each on 17 and 19. That is 1,801 families in all, the smallest of modulus $316,910 = 2 \cdot 5 \cdot 11 \cdot 43 \cdot 67$.

The system. The final system has 618,413 families on the 47 primes up to 211. Every integer from 44 to 160 except 121 is used as a modulus exactly once. The smallest modulus, 44, belongs to the family $\{1 \bmod 4\} \cdot \text{ray}(11, [1, \infty), 0, 1)$ of §3.5. As in [13], the rays are cut off at a finite depth and what remains is covered with the help of an auxiliary prime, here 223. This proves Theorem 1.1; the details are in [10].

9. A REMARK ON 45

We do not know whether these methods reach 45. That would require giving up the modulus 44 as well, which belongs to the family $\{1 \bmod 4\} \cdot \text{ray}(11, [1, \infty), 0, 1)$ of §3.5.

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